

Cryptocurrency Volatility Prediction System High Level Design

1.1 Objective:

The objective of this project is to design and deploy a machine learning-based cryptocurrency volatility prediction system. The system aims to forecast 7-day rolling volatility for various cryptocurrencies to support risk management and informed decision-making by quantifying market "noisiness."

1.2 System Overview : The system follows a structured pipeline to transform raw market data into actionable insights:

- ❖ **Data Ingestion:** Loading historical multi-cryptocurrency records from a CSV.
- ❖ **Data Preprocessing:** Cleaning timestamps and handling missing values.
- ❖ **Feature Engineering:** Creating mathematical "signals" (spreads, liquidity ratios) that correlate with volatility.
- ❖ **Modeling:** Using a Random Forest Regressor to learn patterns between market metrics and future volatility.
- ❖ **Deployment:** A web-based interface built with **Streamlit** for real-time user interaction.

1.3 System Architecture :

- ❖ **Input:** Historical OHLC (Open, High, Low, Close) prices, Volume, and Market Capitalization.
- ❖ **Processing:** A sequential pipeline involving data cleaning, scaling (StandardScaler), and model inference.
- ❖ **Output:** A predicted volatility value (standard deviation) and visual insights (charts and feature importance).

1.4 Technology Stack :

Component	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Model Algorithm	Random Forest Regressor
Serialization	Joblib (for saving .pkl files)
UI & Deployment	Streamlit

1.5 User Interface Design

The Streamlit interface is designed for simplicity:

- **Sidebar Inputs:** Users can adjust market metrics (Daily Return, Spread, Volume) using sliders and number inputs.
- **Prediction Dashboard:** Displays the final volatility score with color-coded warnings (e.g., Green for stable, Red for high risk).
- **Insights Section:** Shows a bar chart of "Feature Importance" to explain which metric is driving the current prediction.